AN ORDINARY PROOF OF THEOREM 3.12 OF CAMERON, GOETHALS, SEIDEL AND SHULT

Abstract. We characterize the sets of lines at angles 90°, 60° and 45° which are contained in the lines of a finite reduced crystallographic Root System. A suitable partition assigns squared lengths 2 and 4 to the initial vectors. In their integer span, we take all vectors of squared length 2 and those of squared length 4 which pair evenly with the span. We prove that this finite set satisfies all Root System axioms. Conversely, crystallographic integrality gives a partition by the parity of the integer logarithm of squared length. We include the empty and nonspanning cases, exact computational checks, and the statement shape of the Lean formalization.

1. Statement and conventions

We will write V for a finite-dimensional real inner product space and $(\mathbf{x}, \mathbf{y})$ for its inner product. Set $q(\mathbf{x}) = (\mathbf{x}, \mathbf{x})$. A *line* is a one-dimensional linear subspace of V; in particular, every line passes through zero. For nonzero $\mathbf{x} \in \ell$ and $\mathbf{y} \in m$, set

$$c(\ell, m) = \frac{(\mathbf{x}, \mathbf{y})^2}{q(\mathbf{x})q(\mathbf{y})}. \tag{1}$$

Replacing either representative by a nonzero multiple multiplies the numerator and denominator by the same square. Thus c is well-defined. For distinct lines, the angles 90°, 60° and 45° mean, respectively, $c = 0$, $c = 1/4$ and $c = 1/2$. These are the unoriented line angles in $[0, 90°]$. We will use their squared cosines throughout, so the argument does not require trigonometry. Entries of matrices and vectors are function values, such as $M(i, j)$ and $\mathbf{x}(i)$; subscripts index families of vectors.

Definition 1.1. [CGSS, Definition 3.9, with the corrections below]. A *Root System* in V is a finite set $\Phi \subseteq V \setminus \{0\}$ which spans V over $\mathbb{R}$ and satisfies:

(1) for every $\mathbf{r}, \mathbf{s} \in \Phi$, $2(\mathbf{r}, \mathbf{s})/q(\mathbf{r}) \in \mathbb{Z}$;
(2) if $\mathbf{r} \in \Phi$, $t \in \mathbb{R}$ and $t\mathbf{r} \in \Phi$, then $t \in \{1, -1\}$; and
(3) for every $\mathbf{r}, \mathbf{s} \in \Phi$,

$$\tau_{\mathbf{r}}(\mathbf{s}) := \mathbf{s} - \frac{2(\mathbf{r}, \mathbf{s})}{q(\mathbf{r})}\mathbf{r} \in \Phi.$$

We will denote its set of lines by $\mathcal{L}(\Phi) = \{\mathbb{R}\mathbf{r} : \mathbf{r} \in \Phi\}$.

This is the finite, reduced, crystallographic convention, with all three qualifiers required. The nonzero condition makes every denominator positive. In the printed condition (b), we read S as Σ. The same definition is used in [R311, Theorems A–C]. We allow $\Phi = \varnothing$ when $V = 0$. Reflection in a root sends that root to its negative, so every root line has both signs; reducedness then says that these are its only two roots.

Theorem 1.2 (Theorem 3.12 of [CGSS], made precise)**.** *Let V be a finite-dimensional real inner product space, and let S be a set of one-dimensional linear subspaces of V. Suppose that for every pair of distinct lines $\ell, m \in S$,*

$$c(\ell, m) \in \{0, 1/4, 1/2\}.$$

The following conditions are equivalent.

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(1) *There is a Root System* Φ *spanning* V *such that* $S \subseteq \mathcal{L}(\Phi)$.
(2) *There are disjoint subsets* $\mathcal{P}, \mathcal{Q}$ *with* $S = \mathcal{P} \cup \mathcal{Q}$ *such that distinct lines in the same part have* $c \in \{0, 1/4\}$, *and lines in different parts have* $c \in \{0, 1/2\}$.

Remark 1.3. Lemma 2.1 proves that S is automatically finite. The set S and either part may be empty, and S need not span V. Allowing an empty part is necessary: a single line lies in a Root System but has no partition into two nonempty parts, and two 60° lines in A_2 must lie in the same part. This is the proposed Implicit hypothesis 8 in `cgss/errata.md`. Allowing a containing Root System in a larger inner product space gives the same condition: the necessary-direction proof uses only the inner products of the selected roots, and the sufficient direction constructs a Root System in V itself.

In the last clause of the printed theorem, "90° and 45°" means "90° or 45°"; this is Error 13, item 3 of the project errata. We prove the equivalence missing in Gap 11. Corollary 3.4 records the stronger construction with squared root lengths in $\{2, 4\}$, both in the span and in V.

Our proof follows route RS3 in `cgss/prerequisites.md`, Sections 6.2 and 13. We will prove both directions directly. The classification in [R311, Theorems A and B] is an allowed prerequisite, but is not needed for this route. The root-angle calculation in its Theorem C is the one we will write out in Lemma 4.1. We will also give the integer-span finiteness argument referred to as LN1 in the prerequisites. Thus we use no unproved completion or root-system classification assertion from [CGSS].

2. Finiteness and the integer span

Lemma 2.1. *Let* V *be a finite-dimensional real inner product space, and let* S *be a set of one-dimensional linear subspaces whose distinct-line squared cosines belong to* $\{0, 1/4, 1/2\}$. *Then* S *is finite.*

Proof. Let U be the set of all unit vectors on the lines of S, including both signs, and let W be its real span. Every line has a unit vector: divide any nonzero vector on it by its positive norm. Thus W is also the span of all the lines. Finite-dimensional linear algebra supplies a basis $\mathbf{b}_1, \dots, \mathbf{b}_d$ of W chosen from U. If U is empty, use the empty basis and $W = 0$. For each $\mathbf{x} \in U$ and each i, either $\mathbf{x}$ is on the line of $\mathbf{b}_i$, when $\mathbf{x} = \pm\mathbf{b}_i$, or the two lines are distinct. In the second case (1) applies. Thus

$$(\mathbf{x}, \mathbf{b}_i) \in \Theta, \qquad \Theta = \{-1, -1/\sqrt{2}, -1/2, 0, 1/2, 1/\sqrt{2}, 1\}.$$

The map $\mathbf{x} \mapsto ((\mathbf{x}, \mathbf{b}_i))_{i=1}^d$ is injective on W. Indeed, if $\mathbf{x}, \mathbf{y}$ have the same image, their difference is orthogonal to every basis vector. By bilinearity it is orthogonal to all of W. Since $\mathbf{x} - \mathbf{y} \in W$, its squared norm is zero, so $\mathbf{x} = \mathbf{y}$. This embeds U into the finite set Θ^d. The map $\mathbf{x} \mapsto \mathbb{R}\mathbf{x}$ maps U onto S, proving finiteness. This argument also covers $d = 0$, when W has no unit vector. □

Lemma 2.2. *Let* $T = \{\mathbf{t}_1, \dots, \mathbf{t}_m\}$ *be a finite set of vectors in a real inner product space. Suppose every* $(\mathbf{t}_i, \mathbf{t}_j)$ *is an integer and every* $q(\mathbf{t}_i)$ *belongs to* $\{2, 4\}$. *Put*

$$L = \left\{ \sum_{i=1}^{m} a(i)\mathbf{t}_i : a(i) \in \mathbb{Z} \right\}, \qquad W = \operatorname{span}_{\mathbb{R}}(T).$$

Then every inner product of elements of L *is an integer, and* $\{\mathbf{x} \in L : q(\mathbf{x}) \le 4\}$ *is finite. More precisely, its cardinality is at most* 9^d, *where* $d = \dim W$. *If* $(\mathbf{t}_i, \mathbf{t}_j)$ *is even for every* j, *then* $(\mathbf{t}_i, \mathbf{x})$ *is even for every* $\mathbf{x} \in L$.

Proof. For integer coefficient functions a, b, bilinearity gives

$$\left(\sum_i a(i)\mathbf{t}_i, \sum_j b(j)\mathbf{t}_j\right) = \sum_{i,j} a(i)b(j)(\mathbf{t}_i, \mathbf{t}_j) \in \mathbb{Z}.$$

The same calculation with $\mathbf{t}_i$ fixed proves the last assertion. No independence of T is needed: any expressions for the vectors give the required integer or even integer.

Choose a basis $\mathbf{b}_1, \ldots, \mathbf{b}_d$ of W from T. For $\mathbf{x} \in L$ with $q(\mathbf{x}) \le 4$, Cauchy–Schwarz gives

$$(\mathbf{x}, \mathbf{b}_i)^2 \le q(\mathbf{x})q(\mathbf{b}_i) \le 16.$$

Thus each $(\mathbf{x}, \mathbf{b}_i)$ is an integer in $\{-4, \ldots, 4\}$. The inner-product coordinate map is injective by the argument in Lemma 2.1. Its target has 9^d elements. If T is empty, then $L = W = \{0\}$, and the assertion gives the correct bound $9^0 = 1$. □

The integral inner products are essential to this argument. We do not infer discreteness merely from finite generation of an additive subgroup of a real vector space. The coordinate proof is the bounded-norm form of LN1, and supplies its proof here.

3. An explicit Root System for the sufficient direction

Lemma 3.1. *Let V be a real inner product space and $\mathbf{r} \in V$ be nonzero. Then the map $\tau_{\mathbf{r}}(\mathbf{x}) = \mathbf{x} - 2(\mathbf{r}, \mathbf{x})\mathbf{r}/q(\mathbf{r})$ is linear, is an involution, preserves squared norms, and satisfies*

$$(\tau_{\mathbf{r}}\mathbf{x}, \mathbf{y}) = (\mathbf{x}, \tau_{\mathbf{r}}\mathbf{y}) \qquad (\mathbf{x}, \mathbf{y} \in V). \tag{2}$$

Also $\tau_{\mathbf{r}}\mathbf{r} = -\mathbf{r}$.

Proof. Linearity follows from bilinearity of the inner product. Direct expansion gives

$$\begin{aligned}
(\mathbf{r}, \tau_{\mathbf{r}}\mathbf{x}) &= -(\mathbf{r}, \mathbf{x}),\\
\tau_{\mathbf{r}}(\tau_{\mathbf{r}}\mathbf{x}) &= \mathbf{x} - \frac{2(\mathbf{r}, \mathbf{x})}{q(\mathbf{r})}\mathbf{r} + \frac{2(\mathbf{r}, \mathbf{x})}{q(\mathbf{r})}\mathbf{r} = \mathbf{x},\\
q(\tau_{\mathbf{r}}\mathbf{x}) &= q(\mathbf{x}) - \frac{4(\mathbf{r}, \mathbf{x})^2}{q(\mathbf{r})} + \frac{4(\mathbf{r}, \mathbf{x})^2}{q(\mathbf{r})} = q(\mathbf{x}),\\
(\tau_{\mathbf{r}}\mathbf{x}, \mathbf{y}) &= (\mathbf{x}, \mathbf{y}) - \frac{2(\mathbf{r}, \mathbf{x})(\mathbf{r}, \mathbf{y})}{q(\mathbf{r})} = (\mathbf{x}, \tau_{\mathbf{r}}\mathbf{y}).
\end{aligned}$$

The denominator is nonzero because $\mathbf{r} \neq 0$. Substitution of $\mathbf{x} = \mathbf{r}$ gives $\tau_{\mathbf{r}}\mathbf{r} = -\mathbf{r}$. □

Proposition 3.2. *Let V be a real inner product space, and let T be a finite set of vectors such that:*

1. *$q(\mathbf{t}) \in \{2, 4\}$ for every $\mathbf{t} \in T$;*
2. *$(\mathbf{t}, \mathbf{u}) \in \mathbb{Z}$ for every $\mathbf{t}, \mathbf{u} \in T$; and*
3. *if $\mathbf{t} \in T$ has $q(\mathbf{t}) = 4$, then $(\mathbf{t}, \mathbf{u}) \in 2\mathbb{Z}$ for every $\mathbf{u} \in T$.*

Let $L = \operatorname{span}_{\mathbb{Z}}(T)$ and $W = \operatorname{span}_{\mathbb{R}}(T)$. The set

$$\begin{aligned}
\Psi(T) = {} & \{\mathbf{x} \in L : q(\mathbf{x}) = 2\}\\
& \cup \{\mathbf{x} \in L : q(\mathbf{x}) = 4 \text{ and } (\mathbf{x}, \mathbf{y}) \in 2\mathbb{Z} \text{ for every } \mathbf{y} \in L\}
\end{aligned} \tag{3}$$

is a Root System in W containing T, with squared root lengths in $\{2, 4\}$.

Proof. Write $\Psi = \Psi(T)$. Hypotheses (1) and (3), together with Lemma 2.2, give $T \subseteq \Psi \subseteq L \subseteq W$. Thus Ψ spans W. Its vectors are nonzero, and it is a subset of the finite set $\{\mathbf{x} \in L : q(\mathbf{x}) \le 4\}$ in that lemma.

Take $\mathbf{r} \in \Psi$. For every $\mathbf{x} \in L$, the reflection coefficient is

$$\frac{2(\mathbf{r},\mathbf{x})}{q(\mathbf{r})} = \begin{cases} (\mathbf{r},\mathbf{x}), & q(\mathbf{r}) = 2; \\ (\mathbf{r},\mathbf{x})/2, & q(\mathbf{r}) = 4. \end{cases} \tag{4}$$

In the first case it is an integer by Lemma 2.2. In the second case it is an integer by the evenness condition in (3). Since L is closed under subtraction and integer multiples, $\tau_{\mathbf{r}}(\mathbf{x}) \in L$. The involution property in Lemma 3.1 then gives $\tau_{\mathbf{r}}(L) = L$.

Take $\mathbf{s} \in \Psi$ and set $\mathbf{t} = \tau_{\mathbf{r}}(\mathbf{s})$. We have just proved $\mathbf{t} \in L$. By norm preservation, $q(\mathbf{t}) = q(\mathbf{s}) \in \{2, 4\}$. If $q(\mathbf{t}) = 4$, then $q(\mathbf{s}) = 4$; for every $\mathbf{x} \in L$, (2) gives

$$(\mathbf{t},\mathbf{x}) = (\mathbf{s}, \tau_{\mathbf{r}}\mathbf{x}) \in 2\mathbb{Z},$$

since $\tau_{\mathbf{r}}\mathbf{x} \in L$. Thus $\mathbf{t} \in \Psi$ in either length case, proving reflection closure.

For $\mathbf{r}, \mathbf{s} \in \Psi$, formula (4), applied to $\mathbf{s} \in L$, proves crystallographic integrality. We will check reducedness explicitly. Suppose $\mathbf{s} = t\mathbf{r}$ with $\mathbf{r}, \mathbf{s} \in \Psi$. Both vectors are nonzero. If their squared lengths are equal, $q(\mathbf{s}) = t^2 q(\mathbf{r})$ gives $t^2 = 1$, so $t = 1$ or $t = -1$. If their squared lengths differ, relabel them and replace t by $1/t$ if needed so that $q(\mathbf{r}) = 2$ and $q(\mathbf{s}) = 4$. Their proportionality then gives

$$t^2 = 2, \qquad t = \frac{(\mathbf{r},\mathbf{s})}{2} \in \mathbb{Q}.$$

The first equation implies $t = \sqrt{2}$ or $t = -\sqrt{2}$, contrary to the irrationality of $\sqrt{2}$. Thus different lengths cannot occur on one line. This proves reducedness.

We have now proved finiteness, nonzero roots, spanning, integrality, reducedness and reflection closure, which are exactly Definition 1.1 in W. If T is empty, then $L = \{0\}$, $\Psi = \varnothing$ and $W = 0$, giving the permitted empty Root System. □

Lemma 3.3. *Let V be a finite-dimensional real inner product space and let $T \subseteq V$ be finite. Suppose that every vector in T has squared length 2 or 4, that all inner products in T are integers, and that each vector of squared length 4 pairs evenly with every vector in T. Put $W = \operatorname{span}_{\mathbb{R}}(T)$, and let $\mathbf{e}_1, \ldots, \mathbf{e}_h$ be an orthonormal basis of $W^{\perp}$. Then*

$$T' = T \cup \{\sqrt{2}\mathbf{e}_i : 1 \le i \le h\}$$

is finite, satisfies the same three conditions, and spans V.

Proof. Each new vector has squared length 2. Its inner products with T and with distinct new vectors are zero; its inner product with itself is 2. Thus all new inner products are integers, and the new pairings involving an old vector of squared length 4 are even. No new vector has squared length 4. This proves the three conditions. The finite union T' spans $W + W^{\perp} = V$ by orthogonal decomposition and $\sqrt{2} \neq 0$. If $W = V$, the new basis is empty; if $W = 0$, it is a basis of V. This also includes $V = 0$. □

Corollary 3.4. *Let V be a finite-dimensional real inner product space and let $S = \mathcal{P} \cup \mathcal{Q}$ be a set of lines, with disjoint parts. Suppose that distinct lines in the same part have squared cosine 0 or $1/4$, and lines in different parts have squared cosine 0 or $1/2$. There is a Root System in the span of S whose lines contain S and whose squared root lengths belong to $\{2, 4\}$. There is also such a Root System spanning V.*

Proof. Lemma 2.1 makes S finite. Choose one vector $\mathbf{t}_\ell$ on each line, of squared length 2 if $\ell \in \mathcal{P}$ and squared length 4 if $\ell \in \mathcal{Q}$. Such a vector is obtained by normalizing any nonzero vector on the line and multiplying by $\sqrt{2}$ or 2. Either sign is permitted. Put $T = \{\mathbf{t}_\ell : \ell \in S\}$. For distinct

lines, (1) and the partition conditions give

parts	$q(\mathbf{t}_\ell)q(\mathbf{t}_m)$	$c(\ell, m)$	$(\mathbf{t}_\ell, \mathbf{t}_m)$
$\mathcal{P}, \mathcal{P}$	4	0 or 1/4	$0,\ 1,\ -1$
$\mathcal{Q}, \mathcal{Q}$	16	0 or 1/4	$0,\ 2,\ -2$
$\mathcal{P}, \mathcal{Q}$	8	0 or 1/2	$0,\ 2,\ -2.$

On the diagonal the inner product is 2 or 4. Thus every inner product is an integer, and every inner product involving a vector of squared length 4 is even. Proposition 3.2 gives the Root System $\Psi(T)$ in the span W of S. Each chosen vector lies in $\Psi(T)$, so its lines contain S. For the ambient assertion, take an orthonormal basis of $W^\perp$ and form T' as in Lemma 3.3. Apply Proposition 3.2 directly to T' in V. Its output $\Psi(T')$ spans V and contains T, so its lines also contain S. Both applications give squared root lengths in $\{2, 4\}$. □

Proof of Theorem 1.2, (2) implies (1). The ambient assertion of Corollary 3.4 gives condition (1). □

4. Root lengths and the necessary direction

Lemma 4.1. [R311, Theorem C, root-angle assertion], in the two cases used here. *Let Φ be a Root System in a real inner product space, and let $\mathbf{r}, \mathbf{s} \in \Phi$ span distinct lines. If their line angle is 60°, then $q(\mathbf{s}) = q(\mathbf{r})$. If it is 45°, then $q(\mathbf{s})/q(\mathbf{r}) \in \{2, 1/2\}$.*

Proof. Put $a = 2(\mathbf{r}, \mathbf{s})/q(\mathbf{r})$ and $b = 2(\mathbf{r}, \mathbf{s})/q(\mathbf{s})$. By crystallographic integrality both are integers. For either angle they are nonzero and have the same sign. Moreover,

$$ab = \frac{4(\mathbf{r}, \mathbf{s})^2}{q(\mathbf{r})q(\mathbf{s})} = 4c(\mathbb{R}\mathbf{r}, \mathbb{R}\mathbf{s}), \qquad \frac{a}{b} = \frac{q(\mathbf{s})}{q(\mathbf{r})}.$$

At 60°, $ab = 1$; the positive integers $|a|, |b|$ must both be 1, so $a/b = 1$. At 45°, $ab = 2$, so their ordered absolute values are $(1, 2)$ or $(2, 1)$, giving $a/b = 1/2$ or 2. This proves the assertions directly from Definition 1.1. □

Proof of Theorem 1.2, (1) implies (2). Let Φ be a Root System in V whose lines contain S. For each $\ell \in S$, choose a root $\mathbf{r}_\ell \in \Phi$ on it and put $\delta(\ell) = q(\mathbf{r}_\ell) > 0$. Reducedness and the presence of both signs show that $\delta(\ell)$ is independent of the chosen root. For $x > 0$, let $\eta(x)$ be the unique integer with

$$2^{\eta(x)} \le x < 2^{\eta(x)+1}. \tag{5}$$

This is $\lfloor \log_2 x \rfloor$, implemented by `Int.log 2 x`. The integer-logarithm bounds in Section 6 supply existence; strict increase of $j \mapsto 2^j$ gives uniqueness. Multiplying (5) by 2 gives $2^{\eta(x)+1} \le 2x < 2^{\eta(x)+2}$. Uniqueness, followed by the same identity with $x/2$ in place of x, gives

$$\eta(2x) = \eta(x) + 1, \qquad \eta(x/2) = \eta(x) - 1. \tag{6}$$

Put in $\mathcal{P}$ the lines for which $\eta(\delta(\ell))$ is even, and in $\mathcal{Q}$ those for which it is odd. Every integer is exactly one of even and odd, including negative integers, so these parts are disjoint and have union S. By Lemma 4.1, a 60° pair has equal squared lengths and equal parities. A 45° pair has squared lengths in ratio 2 or 1/2, so (6) makes its parities opposite. Orthogonal pairs impose no condition. Thus two lines in the same part have angle 90° or 60°, and two lines in different parts have angle 90° or 45°. For $S = \varnothing$ take both parts empty. This proves condition (2) and completes the proof of Theorem 1.2. □

The colouring is defined globally, with no choice of components or paths. Unrelated scales on orthogonal components impose no additional constraint, so a decomposable Root System may have more than two lengths globally. A containing component of type G_2 is also permitted: the angle restriction applies only to the selected lines.

5. Examples and computational checks

5.1. **The role of the parity invariant.** We will first give a small example showing why (3) restricts the vectors of squared length 4. Take independent vectors $\mathbf{b}_1, \mathbf{b}_2, \mathbf{b}_3$ with Gram matrix

$$M = \begin{pmatrix} 2 & -1 & 0 \\ -1 & 2 & 0 \\ 0 & 0 & 2 \end{pmatrix}.$$

Such vectors exist: for an orthonormal basis $\mathbf{e}_1, \mathbf{e}_2, \mathbf{e}_3$, take $\mathbf{b}_1 = \sqrt{2}\mathbf{e}_1$, $\mathbf{b}_2 = -\mathbf{e}_1/\sqrt{2} + \sqrt{3/2}\mathbf{e}_2$ and $\mathbf{b}_3 = \sqrt{2}\mathbf{e}_3$. All initial vectors have squared length 2 and have integral inner products. In their integer span the vector $\mathbf{r} = \mathbf{b}_1 + \mathbf{b}_3$ has squared length 4, but $(\mathbf{r}, \mathbf{b}_2) = -1$. Thus

$$\frac{2(\mathbf{r}, \mathbf{b}_2)}{q(\mathbf{r})} = -\frac{1}{2}, \qquad \tau_{\mathbf{r}}(\mathbf{b}_2) = \mathbf{b}_2 + \frac{1}{2}(\mathbf{b}_1 + \mathbf{b}_3) \notin L.$$

The nonmembership follows from independence of the basis and the noninteger coefficients. Thus taking every vector of squared length 2 or 4 in L would fail. The evenness condition excludes this $\mathbf{r}$ from $\Psi(T)$.

The set $\Psi(T)$ need not be the least reflection-closed set containing T. For two orthogonal seeds of squared length 2, it has eight roots, whereas reflection closure has four. For the usual D_4 seeds, $\Psi(T)$ has 48 roots of type F_4, whereas reflection closure has 24. We will retain reflection closure as a separate computational comparison below; its counts are not counts of $\Psi(T)$.

5.2. **Classical models and the exceptional model.** We will use the coordinate models proved in [R311, Theorem B and the coordinate-model section]. Let $\mathbf{e}_1, \ldots, \mathbf{e}_k$ be the standard orthonormal basis. Their root sets are

$$\begin{aligned} \Phi(\mathrm{B}_k) &= \{\pm\mathbf{e}_i : 1 \le i \le k\} \cup \{\pm\mathbf{e}_i \pm \mathbf{e}_j : i < j\}, \\ \Phi(\mathrm{C}_k) &= \{\pm 2\mathbf{e}_i : 1 \le i \le k\} \cup \{\pm\mathbf{e}_i \pm \mathbf{e}_j : i < j\}, \\ \Phi(\mathrm{F}_4) &= \{\pm\mathbf{e}_i : 1 \le i \le 4\} \cup \{\pm\mathbf{e}_i \pm \mathbf{e}_j : i < j\} \\ &\qquad \cup \left\{ \frac{1}{2} \sum_{i=1}^{4} \epsilon(i)\mathbf{e}_i : \epsilon(i) \in \{1, -1\} \right\}. \end{aligned}$$

In B_k, short roots are on the axes; in C_k, long roots are on the axes. The two line sets coincide, while the length partitions are interchanged. Multiplying all roots of B_k or F_4 by $\sqrt{2}$ gives squared lengths 2 and 4. The C_k model already has that normalization. The allowed squared cosines and the length partitions satisfy Theorem 1.2, by the root-angle calculation or directly from the coordinate lists.

The script `checks/check_thm312.py` uses the following generating seeds before that common normalization:

B_k	$\mathbf{e}_1 - \mathbf{e}_2, \ldots, \mathbf{e}_{k-1} - \mathbf{e}_k, \mathbf{e}_k$
C_k	$\mathbf{e}_1 - \mathbf{e}_2, \ldots, \mathbf{e}_{k-1} - \mathbf{e}_k, 2\mathbf{e}_k$
F_4	$\mathbf{e}_2 - \mathbf{e}_3, \mathbf{e}_3 - \mathbf{e}_4, \mathbf{e}_4, (\mathbf{e}_1 - \mathbf{e}_2 - \mathbf{e}_3 - \mathbf{e}_4)/2.$

It verifies all Root System axioms on the full coordinate sets, computes the partition from line angles, and checks its agreement with the two length classes. Independently, it constructs reflection closure from the displayed seeds with the normalized integer Gram matrix, checking membership in $\Psi(T)$ at every new root. It then compares the entire generated set with the coordinate root

set, in seed coordinates. We check $2 \le k \le 8$ in types B, C, including C_2, and all of F_4. The numbers of generated roots are $2k^2$, $2k^2$ and 48, respectively; their numbers of lines are k^2, k^2 and 24. In rank 2 the angle 60° does not occur in the B, C line set.

5.3. **Containing and noncontaining examples.** The following integer Gram matrices give small sets satisfying the partition condition. The diagonal specifies the assigned squared lengths. Each count is the number of roots after reflection closure; the initial set consists of one vector per matrix row.

Example	Seed Gram matrix	Roots in the closure
Empty set	empty	0
One line	(2) or (4)	2
Two orthogonal lines	$\mathrm{diag}(2,4)$	4
Two 60° lines	$\left(\begin{smallmatrix} 2 & -1 \\ -1 & 2 \end{smallmatrix}\right)$	6
Two 45° lines	$\left(\begin{smallmatrix} 2 & 2 \\ 2 & 4 \end{smallmatrix}\right)$	8
Three lines	$\left(\begin{smallmatrix} 2 & 2 & 1 \\ 2 & 4 & 2 \\ 1 & 2 & 2 \end{smallmatrix}\right)$	18
Three dependent lines	$\left(\begin{smallmatrix} 2 & -1 & -1 \\ -1 & 2 & -1 \\ -1 & -1 & 2 \end{smallmatrix}\right)$	6

For instance, the three-line matrix with 18 roots is realized by $\mathbf{e}_1 + \mathbf{e}_2$, $2\mathbf{e}_1$, $\mathbf{e}_1 + \mathbf{e}_3$ in $\mathbb{R}^3$ and generates C_3. The last matrix is realized by $\mathbf{e}_1 - \mathbf{e}_2$, $\mathbf{e}_2 - \mathbf{e}_3$, $\mathbf{e}_3 - \mathbf{e}_1$; its span is a plane. These include sets which are not initially reflection-closed. The checks also place a two-line 45° configuration in a plane in $\mathbb{R}^4$ and add two orthogonal seeds of squared length 2, as in Lemma 3.3. Reflection closure has 12 roots, while $\Psi(T')$ has 32; the script asserts rank 4 for both outputs. A further check uses $B_2 \oplus 3B_2$. Its four squared lengths are $1, 2, 9, 18$, so this example checks independent component scales.

For noncontainment, put $\gamma = 1/\sqrt{2}$ and consider the two unit Gram matrices

$$M_1 = \begin{pmatrix} 1 & \gamma & \gamma \\ \gamma & 1 & \gamma \\ \gamma & \gamma & 1 \end{pmatrix}, \qquad M_2 = \begin{pmatrix} 1 & 1/2 & \gamma \\ 1/2 & 1 & 1/2 \\ \gamma & 1/2 & 1 \end{pmatrix}.$$

Both are realized by three independent real unit vectors. For M_1, explicit vectors are

$$\mathbf{e}_1, \quad \gamma\mathbf{e}_1 + \gamma\mathbf{e}_2, \quad \gamma\mathbf{e}_1 + (1-\gamma)\mathbf{e}_2 + \sqrt{2\gamma - 1}\mathbf{e}_3.$$

Here $2\gamma - 1 > 0$; their squared norms are 1, and the three inner products are $\gamma, \gamma, \gamma^2 + \gamma(1-\gamma) = \gamma$. For M_2, use

$$\mathbf{e}_1, \quad \tfrac{1}{2}\mathbf{e}_1 + \tfrac{\sqrt{3}}{2}\mathbf{e}_2, \quad \gamma\mathbf{e}_1 + \tfrac{1-\gamma}{\sqrt{3}}\mathbf{e}_2 + \sqrt{\tfrac{1}{2} - \tfrac{(1-\gamma)^2}{3}}\mathbf{e}_3.$$

The last radicand is positive because $0 < 1 - \gamma < 1$. The third vector has squared norm $\gamma^2 + 1/2 = 1$, and its inner products with the first two are γ and $\gamma/2 + (1-\gamma)/2 = 1/2$. Thus the matrices describe actual line sets, not just proposed angles.

In M_1, every pair has angle 45°. A compatible partition would put every pair in different parts, which is impossible on three lines. In M_2, the two 60° pairs force all three lines into the same part, while the 45° pair forces two of them into different parts. Thus neither configuration is contained in a Root System, by the necessary direction. The same obstruction applies in any larger ambient space. The second example also shows why testing only bipartiteness of the graph of 45° pairs is insufficient: its graph has just one edge. One must also enforce equality of parts along 60° pairs.

5.4. **What the computations establish.** Run the checks from the repository root with

```
python cgss/proofs/thm312/checks/check_thm312.py,
```

for a Python 3 with SymPy and NumPy. The script prints its results and writes no files. We ran it with Python 3.12.14, SymPy 1.14.0 and NumPy; its output records the installed versions. Integers, rational numbers and exact algebraic numbers determine every acceptance decision. Numerical eigenvalues are additional diagnostics for the two noncontaining examples, not the criterion for positive definiteness.

The closure calculation stores integer coefficient vectors a in the seeds and uses Ma as the key for a represented vector. This also handles dependent seeds: if M is their Gram matrix, then $Ma = Mb$ says that the represented difference is orthogonal to every seed. It lies in their span, so it is zero. Thus different coefficient expressions of one vector are identified. For each resulting set the script checks nonzero vectors, both signs, spanning, reducedness, integrality, reflection closure and squared lengths. For the normalized seeds it also checks even inner products of long roots with every seed. Evenness against all of L follows by integer linearity, as in the proof.

The explicit set $\Psi(T)$ is checked separately for the independent seed sets of A_1^2, A_1^3, $\mathrm{A}_2 \oplus \mathrm{A}_1$, B_2, C_3, D_4, the ambient-completion example, and the empty set. To enumerate it completely, for a positive definite seed Gram matrix M and an integer coefficient vector a, the script uses

$$q\left(\sum_i a(i)\mathbf{t}_i\right) \le 4 \quad \Longrightarrow \quad a(i)^2 \le 4M^{-1}(i,i).$$

This is Cauchy–Schwarz against the dual basis. The resulting finite coefficient box contains every candidate, so enumeration has a proved bound. The script tests the parity condition and all Root System axioms, and compares the result with reflection closure.

There are exactly $5^3 = 125$ labelled symmetric 3×3 matrices with diagonal 1 and each upper off-diagonal entry in $\{0, 1/2, -1/2, 1/\sqrt{2}, -1/\sqrt{2}\}$. The script checks all of them. Their proper principal minors are positive; the exact determinant decides positive semidefiniteness. The 97 feasible matrices have the following cross-tabulation, which the script asserts and prints:

	With partition	Without partition	Total
Positive definite	65	16	81
Singular	16	0	16
Total	81	16	97

A graph propagation algorithm and exhaustive enumeration of all eight binary colourings agree in every case. For each accepted configuration the script constructs and verifies the reflection closure for both assignments of 2 and 4 to its parts. The 162 resulting closures have sizes $6, 8, 10, 12, 18$, occurring $10, 36, 12, 32, 72$ times, respectively. Independently of those colourings, the script tries all 27 assignments of squared lengths in $\{1, 2, 4\}$ for each of the 97 feasible unit Gram matrices. It checks both ordered crystallographic coefficients for every pair of seeds. Each passing assignment is closed under reflections without assuming the normalized parity condition, and the output is checked against every Root System axiom, with its rank asserted. Some integral assignments have nonreduced closures; these are rejected as Root System candidates. Amongst the 2619 assignments, 2190 fail seed integrality, 24 give nonreduced closures, and 405 give Root Systems. In every configuration, an integral assignment exists if and only if a compatible partition exists, and if and only if an assignment has a verified Root System completion. In particular, all 27 assignments fail seed integrality for each of the 16 configurations without a partition. These are finite obstruction checks independent of the colouring calculation. The script also checks the parity-failure example above.

With random seed 20260928, the script samples 3000 proper subsets of 2 to 9 lines from each of

$$\mathrm{F}_4 \oplus \sqrt{3}\,\mathrm{B}_3, \qquad \mathrm{C}_3 \oplus \mathrm{G}_2 \oplus \tfrac{17}{10}\mathrm{B}_2.$$

It verifies the full containing Root Systems first. The G_2 model is the one in [R311, coordinate-model section], in the plane of $\mathbb{R}^3$ with coordinate sum zero. Rational coordinates with block inner-product weights 3 and 289/100 represent the two rescalings exactly. The 1014 subsets with a 30° pair are counted and skipped; all 4986 remaining subsets are checked by graph propagation and by the parity of the integer logarithm of their actual squared root lengths. Common rescaling of squared lengths by 2^{-4} and 2^{-3} also checks negative logarithms and both parity shifts. The output records the counts and asserts coverage of 45° pairs, multiple components and both length classes of the G_2 component.

These computations check finite instances, including obstructions, dependent seeds and degenerate cases. They neither prove the unbounded-rank theorem nor supply Lean proof terms. Finiteness in all dimensions is proved in Lemma 2.2 and Proposition 3.2. For the normalized seeds, reflection closure stays inside the finite reflection-closed set $\Psi(T)$, which also proves termination of that comparison calculation. The ordinary proof does not use any of the computed counts.

6. Sources and the mathematical interface

We will record the precise division between existing facts and the arguments proved here. Our Mathlib reference [Mathlib] is Mathlib at commit

`082e2d37e8b0463410cdb532e111cd43d5a66174`

(4 September 2026), the version pinned in `lake-manifest.json`. The names below were checked against its source files.

(1) From [CGSS] we use the meaning of a Root System in Definition 3.9 and the statement to be proved in Theorem 3.12, corrected as in Section 1. No theorem of [CGSS], including Theorem 3.10, is required as a premise of our proof. In particular, we do not assume a completion theorem.

(2) From [R311], Theorem C supplies the pairwise root-angle assertion; Lemma 4.1 repeats the needed proof from the integrality axiom. Its Theorem B and its verified coordinate models identify the B, C, F systems used for checks. Theorems A and B are not premises of the equivalence. Thus no root-system classification, or two-length theorem for irreducible systems, is needed in the main argument.

(3) Finite-dimensional linear algebra supplies a basis chosen from a spanning set. In `Mathlib/LinearAlgebra/LinearIndependent/Lemmas.lean`, `exists_linearIndependent` gives an independent subset with the same span over a division ring. The declaration `LinearIndependent.finite` in `Mathlib/LinearAlgebra/Dimension/Finite.lean` makes that subset finite in a finite-dimensional space. We apply this over $\mathbb{R}$ in the two finiteness lemmas. The empty span gives the empty basis. For the line interface, `finrank_eq_one_iff'` in `Mathlib/LinearAlgebra/Dimension/FreeAndStrongRankCondition.lean`, applied to the subtype ℓ, supplies a nonzero vector on a one-dimensional subspace. The declaration `eq_span_singleton_of_mem_of_finrank_eq_one` in `Mathlib/LinearAlgebra/FiniteDimensional/Basic.lean` says that $\ell = \mathrm{span}_{\mathbb{R}}\{\mathbf{r}\}$ for any nonzero $\mathbf{r} \in \ell$. In the same file, `finrank_span_singleton` proves the converse dimension assertion for a nonzero vector. These facts justify the chosen representatives and the definition of `rootLines` below.

(4) Bilinearity, positive definiteness and Cauchy–Schwarz are in `Mathlib/Analysis/InnerProductSpace/Basic.lean`. The declarations `real_inner_self_eq_norm_sq`, `inner_self_eq_zero` and `abs_real_inner_le_norm` apply to real inner product spaces and give exactly the norm facts used above. Reflection identities are expanded in Lemma 3.1, rather than taken from an API with a possibly different reflection convention.

(5) Integer-span induction is supplied by `Submodule.span_induction` and `Submodule.span_induction₂` in `Mathlib/LinearAlgebra/Span/Defs.lean`, with scalars $\mathbb{Z}$. We use their ordinary content, finite integer linear combinations, in Lemma 2.2. Finiteness from an injection into a finite set is `Set.Finite.of_injOn` in `Mathlib/Data/Set/Finite/Basic.lean`. These facts justify both finite-target arguments, without a lattice-discreteness theorem.

(6) The irrationality used for reducedness is `irrational_sqrt_two` in `Mathlib/NumberTheory/Real/Irrational.lean`. Strict increase of integer powers of $2 \in \mathbb{R}$ is `zpow_right_strictMono₀` in `Mathlib/Algebra/Order/GroupWithZero/Basic.lean`, with hypothesis $1 < 2$. In `Mathlib/Data/Int/Log.lean`, `Int.zpow_log_le_self` and `Int.lt_zpow_succ_log_self` give (5) for `Int.log 2 x` when $x > 0$. The equivalent bounds `Int.zpow_le_iff_le_log` and `Int.lt_zpow_iff_log_lt`, applied at $2x > 0$, turn the multiplied inequalities into $\eta(x) + 1 \le \eta(2x) < \eta(x) + 2$. Since these are integers, this proves (6) in Lean without an analytic-logarithm API.

(7) An orthonormal basis of a finite-dimensional inner product space is supplied by `exists_orthonormalBasis`, or the indexed choice `stdOrthonormalBasis`, in `Mathlib/Analysis/InnerProductSpace/PiL2.lean`. We apply it to $W^\perp$ with its induced inner product. Orthogonal decomposition is `Submodule.isCompl_orthogonal` in `Mathlib/Analysis/InnerProductSpace/Projection/Basic.lean`. It assumes `HasOrthogonalProjection`; the instance `Submodule.HasOrthogonalProjection.ofCompleteSpace` supplies this for W, since a finite-dimensional real normed space is complete. Thus its hypotheses are satisfied even when $W = 0$ or $W = V$.

All remaining arithmetic consists of finite sums, the positive integer factor pairs of 1 and 2, or parity of integers. Thus Theorem 1.2 rests only on the listed elementary library facts and the explicit proofs in this document.

7. Lean statement shape

We will keep the settled name `LineSystem.subset_rootSet_iff_bipartition` from `cgss/notation.md`, Section 10. There is a necessary distinction between the namespace and the argument type. The current `LineSystem V`, defined in `LineGraphs/LineSystem/Defs.lean`, already requires all normalized inner products to be in $\{-2, -1, 0, 1, 2\}$. It represents only $60°$ and $90°$ lines. Using an argument of that type would omit all $45°$ cases of this theorem.

We will state the theorem in the `LineSystem` namespace on an unbundled set

$$S : \texttt{Set (Submodule Real V)},$$

with `[NormedAddCommGroup V]`, `[InnerProductSpace Real V]` and `[FiniteDimensional Real V]`. We will require

$$\texttt{hline} : \quad \forall \ell \in S, \quad \texttt{Module.finrank}\ \mathbb{R}\ \ell = 1.$$

The following auxiliary predicates give a precise statement without choosing representatives in its type. They are definitions of the project, in `LineGraphs/Theorem312/Defs.lean`, not declarations of Mathlib.

For subspaces ℓ, m and $a \in \mathbb{R}$, the predicate `LineSystem.HasSquaredCosine` ℓ m a means

$$\forall \mathbf{x} \in \ell\ \forall \mathbf{y} \in m, \qquad (\mathbf{x}, \mathbf{y})^2 = a\, q(\mathbf{x}) q(\mathbf{y}).$$

For lines this is equivalent to (1) having value a: nonzero representatives give the forward implication, and multiplying representatives gives the reverse implication, including zero vectors. The division-free form avoids nonzero proofs in the predicate itself. Define `LineSystem.HasAllowedAngles` S to mean

$$\forall \ell, m \in S,\ \ell \neq m \implies \exists a \in \{0, 1/4, 1/2\}, \quad \texttt{HasSquaredCosine}\ \ell\ m\ a.$$

Define `LineSystem.HasBipartition` S to mean that there is a function $b : S \to \texttt{Bool}$ such that for distinct $\ell, m \in S$,

$$\begin{cases} \texttt{HasSquaredCosine}\ \ell\ m\ 0\ \vee \texttt{HasSquaredCosine}\ \ell\ m\ (1/4), & b(\ell) = b(m); \\ \texttt{HasSquaredCosine}\ \ell\ m\ 0\ \vee \texttt{HasSquaredCosine}\ \ell\ m\ (1/2), & b(\ell) \neq b(m). \end{cases}$$

The two fibres of b are exactly the parts of condition (2), and neither fibre is required to be nonempty.

The predicate `IsRootSet` on Φ : `Set V` contains the six requirements of Definition 1.1: finiteness, nonzero vectors, real span equal to the top submodule, crystallographic integrality, reducedness and reflection closure. Define `LineSystem.rootLines` Φ by

$$\{\ell : \exists \mathbf{r} \in \Phi, \quad \ell = \mathrm{Submodule.span}_{\mathbb{R}}\{\mathbf{r}\}\}.$$

The nonzero-root field ensures that these are lines whenever `IsRootSet` Φ holds. In particular the statement is line containment, not containment of a particular normalization of root vectors.

With these definitions the declaration has the following conclusion, under the displayed typeclasses and hypotheses `hline` and `hangle : HasAllowedAngles S`:

$$\big(\exists \Phi : \texttt{Set V}, \quad \texttt{IsRootSet}\ \Phi\ \wedge\ S \subseteq \texttt{rootLines}\ \Phi\big) \quad \Longleftrightarrow \quad \texttt{HasBipartition}\ S.$$

The theorem's full name is `LineSystem.subset_rootSet_iff_bipartition`. Finiteness of S is derived from `hline` and `hangle`, as in Lemma 2.1; a finite-set wrapper can expose it explicitly. Spanning and nonemptiness of S are not hypotheses. The docstring of the theorem records these conventions and the errata pointers:

> Theorem 3.12 of [CGSS]. For one-dimensional subspaces with squared cosines 0, 1/4 or 1/2, root-line containment is equivalent to a compatible bipartition. The set and either colour fibre may be empty (Implicit hypothesis 8); finiteness is derived and no spanning hypothesis is imposed on the lines. The containing root set spans the given ambient space. The cross-part condition reads "or" (Error 13, item 3); the explicit construction fills Gap 11.

For the construction, use `Submodule.span Int T` for L and define `LineSystem.integralSpanRoots` T by the set-builder (3). The squared-length-4 condition quantifies over every element of this fixed L. Use the ambient version of Proposition 3.2, with the extra hypothesis $\mathrm{span}_{\mathbb{R}}(T) = V$. First form T' by Lemma 3.3, using `stdOrthonormalBasis` on $W^{\perp}$ and viewing its vectors in V. Then apply the proposition to T' directly. This proof uses no `IsRootSet` on the subtype W and no extension lemma for a union of Root Systems. The span assertion of Corollary 3.4 can be formalized separately by restricting the same construction to W. The helper theorems in the `LineSystem` namespace are `finite_of_hasAllowedAngles`, `isRootSet_integralSpanRoots` and `exists_rootSet_of_hasBipartition`, following the conclusion rule in the notation file.

For the necessary direction, set the Boolean colour to `decide (Int.emod (Int.log 2 (inner r r)) 2 = 1)` for the chosen root on a line. Equations (5) and (6) give the argument without components or path choices.

The namespace follows rule 2 of `cgss/notation.md`, Section 10, which places line-system results in `LineSystem`; the argument is the unbundled set S, not a term of the structure `LineSystem V`. The notation file records this shape and the five definitions `HasSquaredCosine`, `HasAllowedAngles`, `HasBipartition`, `rootLines` and `integralSpanRoots`.

For a bridge to Section 2, if R : `LineSystem V`, then `rootLines` $R.\Phi$ satisfies `HasAllowedAngles` and `HasBipartition`, with the constant Boolean colouring. Indeed, its vectors have squared length 2; inner products ± 2 give equal lines, since the squared norm of their difference or sum is zero. Distinct lines thus have inner product 0 or ± 1 on these representatives, giving squared cosine 0 or 1/4. Equation (1) transfers this to all representatives. The bridge lemmas are `LineSystem.hasAllowedAngles_rootLines` and `LineSystem.hasBipartition_rootLines`.

No bridge to `RootPairing` is needed to prove this statement. If the bridge proved in [R311] is used afterwards, Mathlib's pairing has the squared length of the *second* root in the denominator:

$$p(\mathbf{r}, \mathbf{s}) = \frac{2(\mathbf{r}, \mathbf{s})}{q(\mathbf{s})}.$$

Definition 1.1 uses the first root as the reflecting root, so the bridge interchanges the arguments. Integrality for all ordered pairs provides both conventions. The Root System produced here is reduced and crystallographic and spans its stated ambient type, exactly as required by the bridge in [R311].

There is no unresolved mathematical step in the ordinary proof. The structured version of these proofs is `cgss/proofs/thm312/thm312-lamport.tex`, and the Lean formalization, in `LineGraphs/Theorem312/`, proves `LineSystem.subset_rootSet_iff_bipartition` with the helper definitions and theorems above; it uses no computational check as a mathematical premise.

References

[CGSS] P. J. Cameron, J. M. Goethals, J. J. Seidel and E. E. Shult, *Line graphs, root systems, and elliptic geometry*, J. Algebra **43** (1976), 305–327. Definition 3.9 and Theorem 3.12, pp. 314–315; see also `cgss/errata.md`, Error 13, item 3 and Gap 11.

[R311] *The classification behind Remark 3.11 of [CGSS]*, project manuscript `cgss/proofs/remark311/remark311.tex`, Theorems A–C and the coordinate-model and Mathlib-interface sections.

[Mathlib] The mathlib community, *Mathlib*, commit `082e2d37e8b0463410cdb532e111cd43d5a66174` (4 September 2026), the version pinned in `lake-manifest.json`. The specific source files and declarations used here are listed in Section 6.